PLANNING FOR TRUCK FLEET SIZE IN THE PRESENCE OF A COMMON-CARRIER OPTION

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ABSTRACT

This paper considers a class of network optimization problems in which certain directed arcs must be covered by a set of cycles. Our study was motivated by a distribution planning problem of a commercial firm that had to make deliveries over several origin-destination pairs (directed arcs) and that could service any demand arc by using a vehicle in its own fleet or by paying a common carrier. The problem is to determine an optimal fleet size and the resulting vehicle routes while satisfying maximum route-time restrictions. We formulate the problem, describe some approximate solution strategies, and discuss important implementation issues.

Subject Areas: Logistics and Distribution, Heuristics, and Networks.

INTRODUCTION

National and international industries involved in the production and distribution of large volumes of commodities typically incur enormous shipping expenses. A firm usually has the option of either paying a common carrier to transport its goods or leasing or buying its own fleet of trucks to carry out the distribution. This paper describes the modeling and solution of the distribution problem of a large chemical firm in moving specified demands between origins and destinations distributed widely over most of the continental United States and parts of Canada. This particular firm found it economical to service part of the demand with its own leased fleet and to engage a common carrier to service the remaining portion.

This paper addresses the following problems faced by the firm:

(a) to determine the size of the leased fleet,
(b) to determine which demand trips should be serviced by the leased fleet and which should be servied by common carrier,
(c) to determine the routes for the vehicles in the leased fleet.

Of these, the first problem formed our primary focus, but the firm insisted upon a detailed presentation of the routes before the overall results were accepted, thus placing an equal emphasis on the second and third problems.

The three problems cited are neither atypical to industry nor unique to the firm we studied. Their increasing importance is highlighted due to the present escalation of fuel costs and the prospect of deregulation within the trucking industry. These key factors make it essential that fleet size be reinvestigated on a regular basis.

The distribution/routing problem under study had a structure that we had not encountered previously. The demands were specified in terms of trailer trips per week between origin-destination pairs. Moreover, a load on a trailer could not be split among destination locations and the capacity of a tractor was one trailer. As such, a typical route would behave as follows:

> Leave domicile, pick up a trailer, deliver a trailer, deadhead (nonproductive travel), pick up a trailer, deliver a trailer, pick up a trailer, deliver a trailer, . . ., deliver a trailer, return to domicile.

Because of this structure, the problem lends itself to two different formulations. On the one hand, by considering origins and destinations as nodes, the problem may be viewed as a routing problem with demand on certain network *arcs* (see [1] and [2]). On the other hand, by considering origin-destination (O-D) pairs as nodes, the problem may be stated as a routing problem with demands at certain *nodes*. The latter approach allows the problem to be conceptualized as a standard vehicle-routing problem [3] [4] [5] [9] where one of the vehicles, which represents the common carrier, has significantly different cost and operating characteristics than the others. In order to exploit the special structure of this problem in the construction of an algorithm, we provide an arc-routing formulation whose objective function is to maximize the savings in common-carrier cost resulting from the use of leased vehicles. This is one of the few routing problems that we have encountered that can be formulated as either a node-routing or an arc-routing problem and for which an equally effective solution procedure can be designed using either formulation.

The plan of this paper is as follows. The second section contains the underlying modeling assumptions and data requirements. The third section provides a mathematical programming formulation of the problem. The fourth section describes three solution procedures that are based on both the node- and arc-routing views of the problem. The first two procedures initially generate a large route for a single vehicle with unlimited capacity and then partition this route into smaller routes that can be traversed by real vehicles (a variant of the "route-first" approach described in [1]). The third procedure is a "greedy" insertion procedure (similar to the insertion schemes discussed in [8]). This approach generated the best results in our computational experience. Consequently, its solutions were used by the chemical firm in deciding on a leased-fleet size.

The fifth section examines the results of the model in detail along with certain practical implementation issues. The concluding section discusses generalizations of the model and suggests possible refinements to improve the efficiency of the algorithms developed.

PROBLEM DEFINITION

The chemical firm's requirements were stated in terms of the number of truckloads of goods (which we called trips) that must be shipped from a particular origin to a particular destination per week. Origins and destinations were of several types including warehouses, distribution centers, and customer locations. The weekly demands for such O-D pairs ranged between .25 and 10 trips per week and averaged about 2 trips per week across all O-D pairs. We assumed that trip demands were constant and known inputs into our model. In actuality, however, the demands fluctuated slightly from week to week. The operational practice of the firm was to respond to such fluctuations by determining new routes at the beginning of each week based on trip demands for that week. Our assumption of constant demands was more appropriate to the tactical problem of determining the leased-fleet size, which formed our primary focus, than to the operational problem of generating weekly vehicle routes.

There was a variety of loosely stated restrictions on the timing of deliveries. In particular, some facilities could accommodate deliveries twenty-four hours a day while others could only accept deliveries during certain periods of the day. Furthermore, if a particular O-D pair had a high weekly demand, then it was desirable that shipments be spread out over the week.

We formulated and solved the routing problem without explicitly considering the timing restrictions. We found that the solutions obtained rarely violated these constraints; and, when they did, minor modifications to the solution produced a feasible solution.

The leased trucks were two-man tractor trailers that could stay on the road for 120 continuous hours per week. Thus, 120 hours became the upper bound on the length of the route for any leased vehicle. Trip durations varied between 3 hours and 50 hours with most trips lasting less than 15 hours. Trucks could be housed at any one of three domiciles. Consequently, the problem had a "multi-depot" nature in that the choice of a domicile had to be included in the specification of a route. The common-carrier tariff structure was readily available through ICC publications (since most movements were interstate) and thus was independent of the number of trips required. The leased truck times and costs included portions that varied with mileage and fixed portions associated with pick-up and delivery times at the various origins and destinations. The speed of deadheading an arc was greater than the speed of traversing an arc with a full load.

A small routing example is presented in Figure 1A. The nodes represent facilities, and the directed arcs represent origin-destination pairs. The number of trips required is given in parentheses. The total number of O-D pairs is 6, and the total number of weekly trips is 9. A possible solution is shown in Figure 1B. Two vehicle routes are shown with broken lines, and the numbers in brackets specify the trip demands serviced by common carrier. The first route traverses the path (domicile, E, C, A, B, D, C, D, C, domicile). The second route traverses the path (domicile, I, J, G, H, domicile). The common-carrier services one demand trip each over (E, F), (A, B), and (G, H).

The data requirements of the model are straightforward. For a leased vehicle servicing a demand arc (i,j),

t_{ij} = load time at i + travel time + unload time at j,

c_{ij} = crew wait cost at i + vehicle and crew travel cost + crew wait cost at j,

and for a common carrier servicing a demand arc (i,j)

r_{ij} = the tariff rate.

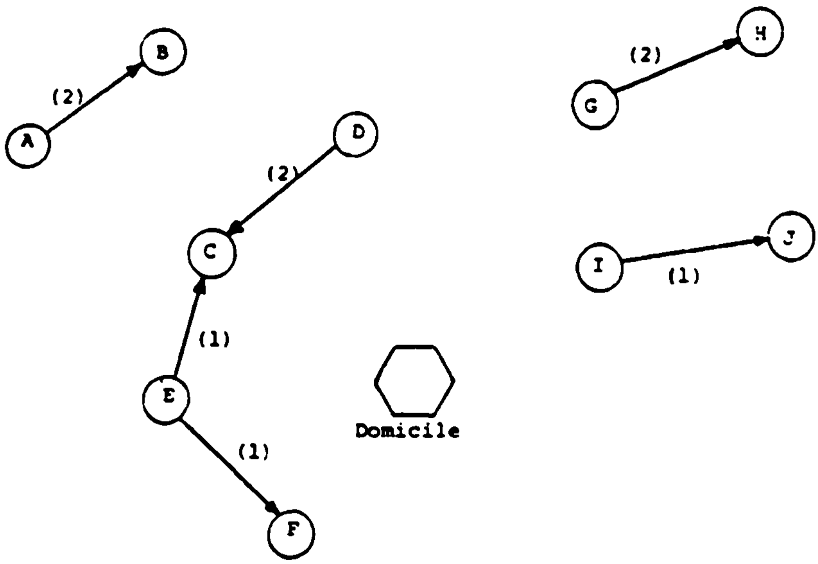
For a leased vehicle traversing a deadhead arc (i,j)

s_{ij} = travel time, and

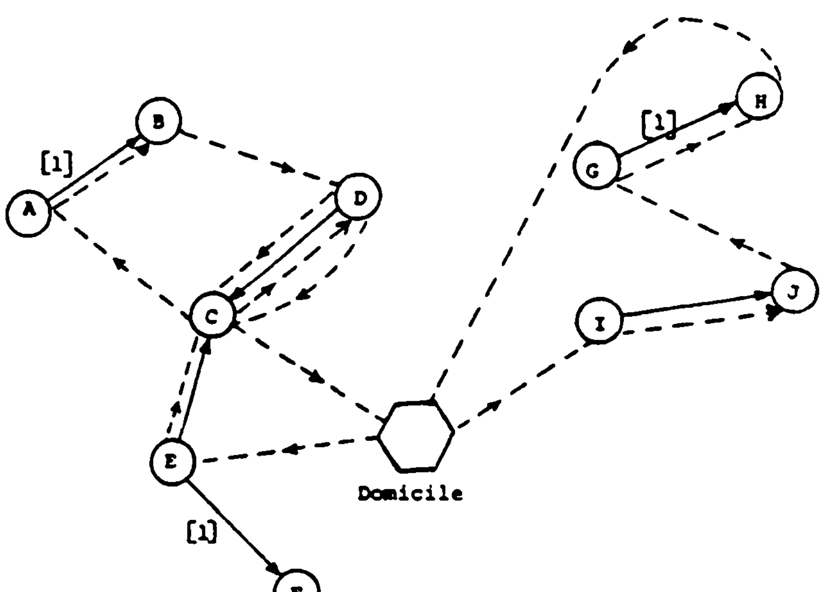
b_{ij} = vehicle and crew travel cost.

FIGURE 1
A Sample Routing Problem and a Solution

A. Routing Problem



B. Possible Solution



Also required are: the number of trips, d_{ij}, required per week from i to j; a fixed weekly vehicle leasing charge, $\bar{c}$; and a maximum weekly route time for a leased vehicle, T.

FORMULATION

This section provides a precise specification of the routing problem confronting us. The next section describes several algorithms for solving the formulated problem to yield a near-optimal solution.

We considered two possible formulation strategies. First, we may envision each demand trip as a node. Such nodes are joined by directed arcs that represent deadheading from the end of one trip to the beginning of the next. Alternatively, we may consider each facility a node. This view gives rise to two types of directed arcs:

(A) demand arcs between each origin and its destinations representing trips, and

(B) deadhead arcs from each destination node to each origin node.

Traversing arcs of Types (A) and (B) then corresponds, respectively, to servicing a trip and to deadheading from the end of one trip to the beginning of another. The particular data we worked from had the property that many O-D pairs shared the same origin, and, in a few cases, origins of certain trips coincided with destinations for others. We chose the second formulation approach since it better captured the structure of the problem and required fewer constraints and variables.

We now present some notation from the mathematical program which we denote by P. Let

A = set of demand arcs,

E = set of deadhead arcs,

DOM = set of domicile nodes,

y_{ijk} = the number of trips from i to j serviced by vehicle k for $(i,j) \in A$, and

x_{ijk} = the number of deadhead runs from i to j made by vehicle k for $(i,j) \in E$.

The objective function is to minimize cost:

$$P\text{: Min} \sum_{(i,j)\in A} \{d_{ij} - \sum_k y_{ijk}\} r_{ij} + \sum_k \sum_{(i,j)\in A} c_{ij} y_{ijk} + \sum_k \sum_{(i,j)\in E} b_{ij} x_{ijk}. \tag{1}$$

The constraints are:

$$\sum_{(i,j)\in A} t_{ij} y_{ijk} + \sum_{(i,j)\in E} s_{ij} x_{ijk} \le T, \qquad \text{for all } k, \tag{2}$$

$$\sum_k y_{ijk} \le d_{ij}, \qquad \text{for all } (i,j)\in A, \tag{3}$$

$$\{\sum_{j:(i,j)\in A} y_{ijk} + \sum_{j:(i,j)\in E} x_{ijk}\} - \{\sum_{j:(j,i)\in A} y_{jik} + \sum_{j:(j,i)\in E} x_{jik}\} = 0,$$

$$\text{for all } i \text{ and } k, \tag{4}$$

$$\sum_{j:(i,j)\in A\cup E} f_{ijk} - \sum_{j:(i,j)\in A\cup E} f_{jik} = \sum_{j:(i,j)\in A} y_{ijk} + \sum_{j:(i,j)\in E} x_{ijk},$$

$$\text{for all } i \notin DOM \text{ and all } k, \tag{5}$$

$$f_{ijk} \leq u(y_{ijk} + x_{ijk}), \qquad \text{for all } (i,j)\in A\cup E \text{ and all } k, \tag{6}$$

$$f_{ijk} \geq 0, \qquad \text{for all } i, j, \text{ and } k, \tag{7}$$

$$x_{ijk}, y_{ijk} \geq 0 \text{ and integer}, \qquad \text{for all } i, j, \text{ and } k. \tag{8}$$

Constraint set (2) ensures that no vehicle exceeds its time limitation. Constraint set (3) limits the number of times an O-D pair is serviced by the leased vehicles. Conservation of flow is guaranteed by (4). Subtours are eliminated by (5), (6), and (7). To see this, f_{ijk} is envisioned as a flow variable that can take on positive values only if $x_{ijk} > 0$ or if $y_{ijk} > 0$ by Constraint (6). If we consider a subtour that does not include a domicile and is not connected to a domicile and we sum Equation (5) over all arcs in the subtour, we obtain zero on the left side of the equality and some positive number on the right side. Constraints (5) and (6) therefore prohibit this configuration. The constant u in (6) should be suitably large to render (6) redundant when $y_{ijk} + x_{ijk} > 0$. A fuller discussion and illustration of the subtour-breaking constraints appear in Appendix A.

SOLUTION ALGORITHMS

Three solution algorithms were proposed. The first two follow a "route-first, cluster-second" strategy and the third uses a "greedy" insertion procedure. All three algorithms use the concept of route savings, defined as that portion of the objective function associated with the savings yielded by a particular route k over the common-carrier cost. It is given by

$$\sum_{(i,j)\in A(k)} (r_{ij} - c_{ij}) - \sum_{(i,j)\in E(k)} b_{ij}, \tag{9}$$

where $A(k)$ and $E(k)$ refer to the sets of demand and deadhead arcs on route k, respectively. If we maximize the summation over all k of (9), the result is equivalent to the original objective function (Equation (1)).

Route-First, Cluster-Second Approaches

We devised two specific route-first, cluster-second procedures. Both consider a related single-vehicle routing problem. The single-vehicle problem denoted by

P' seeks to find a single route that covers all demand. This problem is obtained from P by deleting Constraint set (2) and by changing the inequality in (3) to an equality.

The resulting problem is a directed rural postman problem [10] [11]. Alternatively, when viewed in terms of the "node" formulation discussed in the third section of this paper, this problem is a traveling salesman problem (TSP) [8]. Algorithm 1A initially solves a relaxation of the rural postman problem and Algorithm 1B initially solves, approximately, the induced TSP.

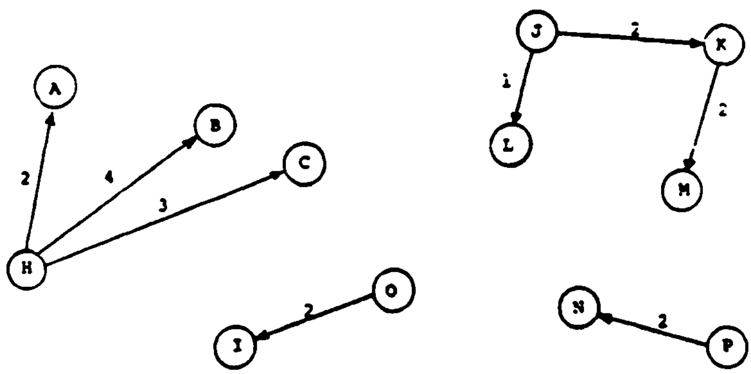
The first procedure (Algorithm 1A) considers a problem P'' obtained from P' by dropping the subtour-elimination constraints of (5), (6), and (7). It solves this relaxation by constructing an equivalent transportation problem (see Figure 2). The demand points in the transportation problem correspond to network nodes i, for which $\sum_j d_{ij} - \sum_j d_{ji} > 0$, with the demand being equal to the value of the left-hand side of this inequality. Similarly, the supply points are those network nodes, i, for which $\sum_j d_{ji} - \sum_j d_{ij} > 0$, with the supply again being the value of the left-hand side of this inequality. In Figures 2A and 2B, H has a demand of 9 and B has a supply of 4 units. The cost of transporting a unit of supply from supply point i to demand point j is b_{ij} where b_{ij} is the cost of the corresponding deadhead arc. The solution to this transportation problem gives the minimum-cost augmentation of deadhead arcs which is a least-cost set of deadhead arcs that guarantees that the in-degree equals the out-degree for each node in the network (see [1] for amplification). In Figure 2C, these arcs are indicated by the broken lines. If the augmented network admits directed paths between all nodes i and j, then a Eulerian circuit (a circuit covering all arcs) exists and can be found easily. This circuit is the solution to the rural postman problem (see [2] and [6]). In general, however, there is not a directed path from each node to every other node and the network contains several strongly connected components. A directed network is strongly connected if paths in both directions exist between any pair of nodes. In Figure 2C, two strongly connected components result. The solutions we obtained contain between 4 and 6 strongly connected components.

Algorithm 1A partitions the set of strongly connected components obtained from the solution of P'' into a set of routes whose travel times do not exceed $T = 120$. The details of the algorithm are not given here.

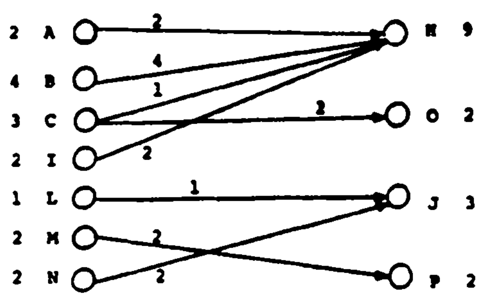
Algorithm 1B solves the rural postman problem by considering each demand trip as a node, forming the network described in the "node" formulation of the third section of the paper, and then computing a TSP tour in this directed network (Figure 3A). The TSP tour is used to generate a set of candidate routes with travel time less than or equal to $T = 120$. The candidate routes are obtained by taking sets of consecutive nodes on the TSP tour and then connecting the endpoints of each path to the closest domicile (see Figure 3B). Given K leased vehicles, the problem is to choose a maximum savings set of K candidate routes no two of which cover the same node.

FIGURE 2
Solution Steps for Algorithm 1A—
A Route-First, Cluster-Second Strategy in Terms of a Rural Postman Problem

A. Demand Arcs for Problem *P*"



B. Transportation Problem Solution



C. Rural Postman Network
(with 2 strongly connected components)

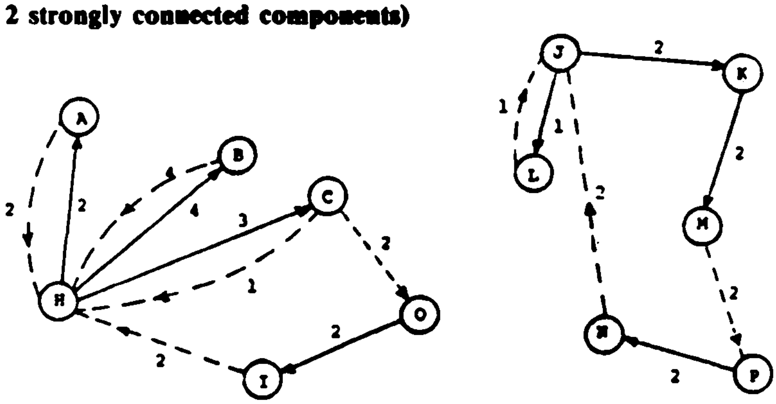
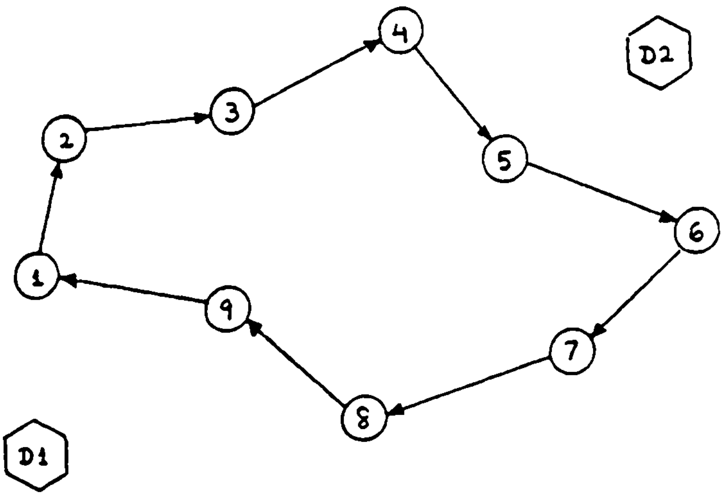
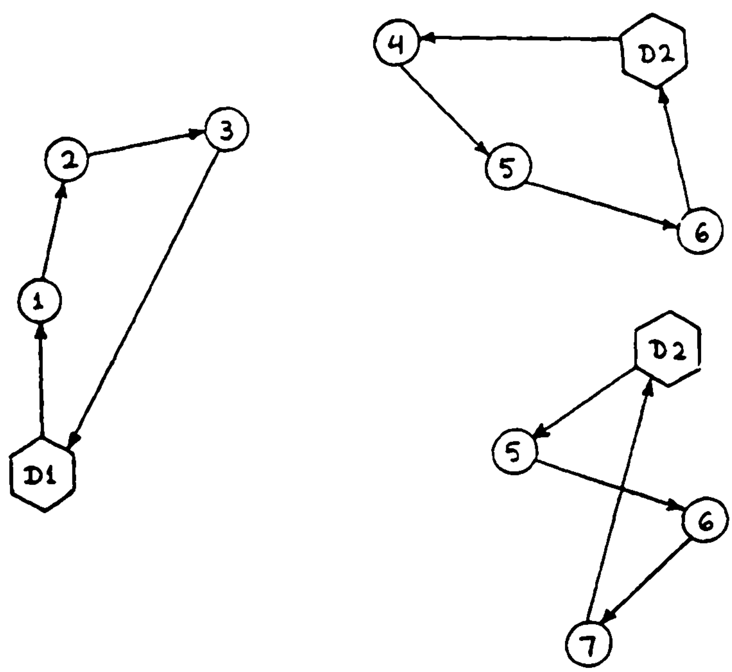


FIGURE 3
Solution Steps for Algorithm 1B—
A Route-First, Cluster-Second Strategy in Terms of a Traveling Salesman Problem

A. TSP Tour and Two Domiciles



B. Some Candidate Routes



Although this approach assumes K to be a prespecified number of leased vehicles, we can repeat Algorithm 1B for different values of K. Algorithm 1B is described in more detail in Appendix B.

A "Greedy" Insertion Approach

Insertion procedures build routes by adding nodes or arcs to them, one at a time, until no additional feasible insertions are possible or profitable. The insertion procedure we considered constructs a route in an iterative fashion by adding demand arcs to it one at a time. It uses the ratio of the marginal increase in savings divided by the marginal increase in route time as a criterion for adding a demand arc. This "bang-for-buck" ratio is analogous to a similar ratio used in the well-known greedy algorithm for the knapsack problem [12].

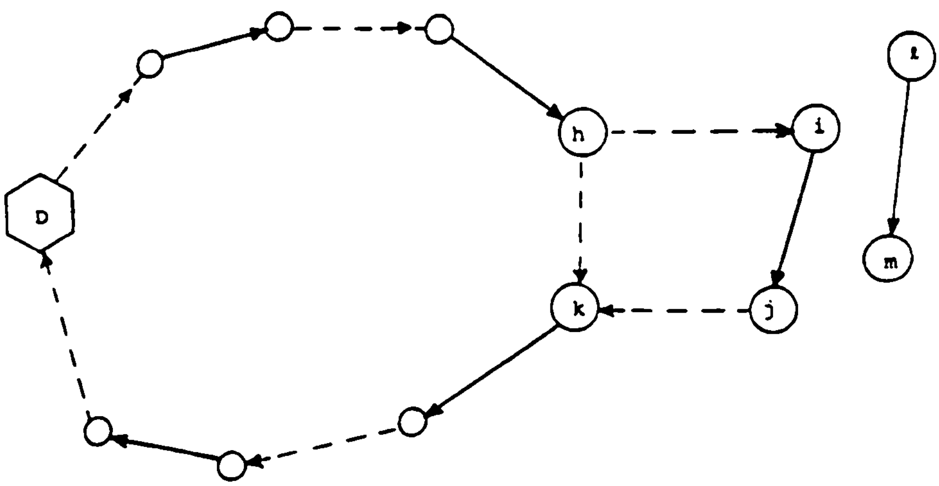
Figure 4 illustrates the idea behind this algorithm and gives the associated mathematical formulas. The steps of the algorithm are stated below.

Algorithm 2:

Step 1: Select an initial route.

Step 2: While a feasible insertion exists (a) perform the insertion with the best "bang-for-buck" ratio and (b) decrement the demand on the arc chosen.

FIGURE 4
Solution Step for Algorithm 2—
A Greedy Insertion Approach



$$\Delta S = (r_{ij} - c_{ij}) - b_{hi} - b_{jk} + b_{hk}$$

$$\Delta T = t_{ij} + s_{hi} + s_{jk} - s_{hk}$$

"bang-for-buck" ratio $= \Delta S / \Delta T$

Step 3: Add the route to the route list. If all demands are zero, go to Step 4; otherwise, go to Step 1.

Step 4: Order the routes by decreasing route savings. The first K routes are the solution to the K-vehicle problem. Other demands will be serviced by common carrier.

Computational Experience

All three algorithms were tested in order to determine which would yield the best results in terms of solution value and computational efficiency. Computation time turned out to be a nonsignificant issue since for the problem size under consideration (150-200 O-D pairs, 400-800 total trips) all procedures ran within five minutes, an acceptable amount of time, on an IBM 370/148 using FORTRAN-IV.

In preliminary tests for relatively small values of K, the total savings obtained by Algorithm 2 were 10 to 15 percent higher than those of Algorithm 1A, which in turn outperformed Algorithm 1B by about the same margin. Consequently, we decided to implement fully only Algorithm 2 and Algorithm 1A. Algorithm 2 produced the best results for small- to moderately-sized values of K (i.e., less than 60 percent of the demand handled by the leased fleet), and Algorithm 1A produced better results for large values of K. These results were anticipated since Algorithms 1A and 1B seek to service all demand trips at minimal total cost whereas Algorithm 2 seeks to extract and service only the most profitable demand trips. The particular problems considered had optimal solutions at relatively small values of K, and consequently the solutions of interest to the chemical firm were derived from Algorithm 2.

Since Algorithm 2 produced the best results, certain computational issues that arose in its implementation will be discussed.

Step 1 requires a method for finding an initial route. Although we included the capability for inserting a set of initial routes, a very simple initialization procedure proved more effective. For each domicile, a dummy route consisting of an arc from the domicile back to itself was used. The best insertion is found for each of these routes, and then the algorithm is run starting with the route that has the greatest savings-to-time ratio.

Step 2 requires the greatest computational effort. Our implementation iteratively scanned the list of demand arcs, found the insertion yielding the largest "bang-for-buck" ratio for each arc, and then computed the best insertion over all arcs. Computing the best ratio for each arc can be done particularly efficiently since in most cases the ratio does not change from one iteration to the next. Referring to Figure 4, suppose that demand arc (i,j) has just been added to the partial route and we wish to compute the new best ratio for demand arc (ℓ,m). Then we can take advantage of the following.

Observation. After arc (i,j) is added, the "bang-for-buck" ratio changes for arc (ℓ,m) if and only if one of the following occurs: (1) the new best insertion is between h and i or j and k, (2) the previous best insertion was between h and k, or (3) the previous best insertion is no longer feasible. Thus, no update is necessary unless either Case (1), (2), or (3) occurs. For Case (1), a single calculation is required. Consequently, only in Cases (2) and (3) must all possible insertions along the current partial route be computed and compared.

FINDINGS AND INTERPRETATION OF RESULTS

As mentioned in the second section of this paper, there was a variety of loosely stated constraints on timing of deliveries. In some cases, the solutions generated satisfied these constraints, and in other cases minor modifications were required. Two mechanisms for modifying the solutions were used. First, a capability was included in each of the algorithms for ensuring that repeated trips between a particular O-D pair were spread out in time over each route. In other words, an attempt was made to replace a route such as

domicile, A, B, A, B, A, B, C, D, C, D, domicile

with

domicile, A, B, C, D, A, B, C, D, A, B, domicile.

Second, sets of consecutive trips along a route were sometimes interchanged in order to satisfy time-window restrictions. We were able to eliminate all infeasibilities in this manner, and these modifications never increased the total solution cost by more than 2 percent.

Previous fleet operations had set aside certain trips for the common carrier. A solution obtained by Algorithm 2 in which any demand trip could be serviced by a leased vehicle was compared to a solution in which the leased fleet was prohibited from servicing the demand trips currently allocated to the common carrier. The first solution had a total savings of over $30,000 per week as compared to the second solution's savings of $15,000 per week.

Since the primary purpose of this study was to determine the *size* of the leased fleet, we were more concerned with broad strategic planning issues than day-to-day operational vehicle routing. As the firm was about to lock itself into a long-term leasing contract, it felt the need for some protection against the possibly adverse effects of long-term changes in trip demands. In particular, management wanted to avoid leasing too large a fleet lest total demand decrease. Hence, management was willing to have some deliveries handled by common carrier even though, in the short run, it might have been more profitable to lease additional vehicles, hire more drivers, and service these extra deliveries with the firm's own fleet.

To consider explicitly the company's attitude towards risk, a concept similar to the marginal rate of return was used to identify highly cost-effective routes to be serviced by the leased fleet. Let $C(K)$ denote the cost of a heuristic solution using K vehicles—that is, the leased-fleet cost associated with the routing heuristic solution for K vehicles plus the cost of servicing all remaining demand by common carrier. Let $S(K)$ be the *savings* associated with a K-vehicle fleet—that is $S(K)$ is the difference between the cost of servicing all demand by common carrier and

$C(K)$. In addition, let $D(K)$ be the leased-fleet costs associated with the routing heuristic solution for K vehicles. Therefore, $C(K)$ equals $D(K)$ plus the cost of servicing by common carrier those demands not handled by the leased fleet.

The marginal savings rate for k vehicles is then given by

$$M(K) = (S(K) - S(K-1))/D(K) - D(K-1)). \tag{10}$$

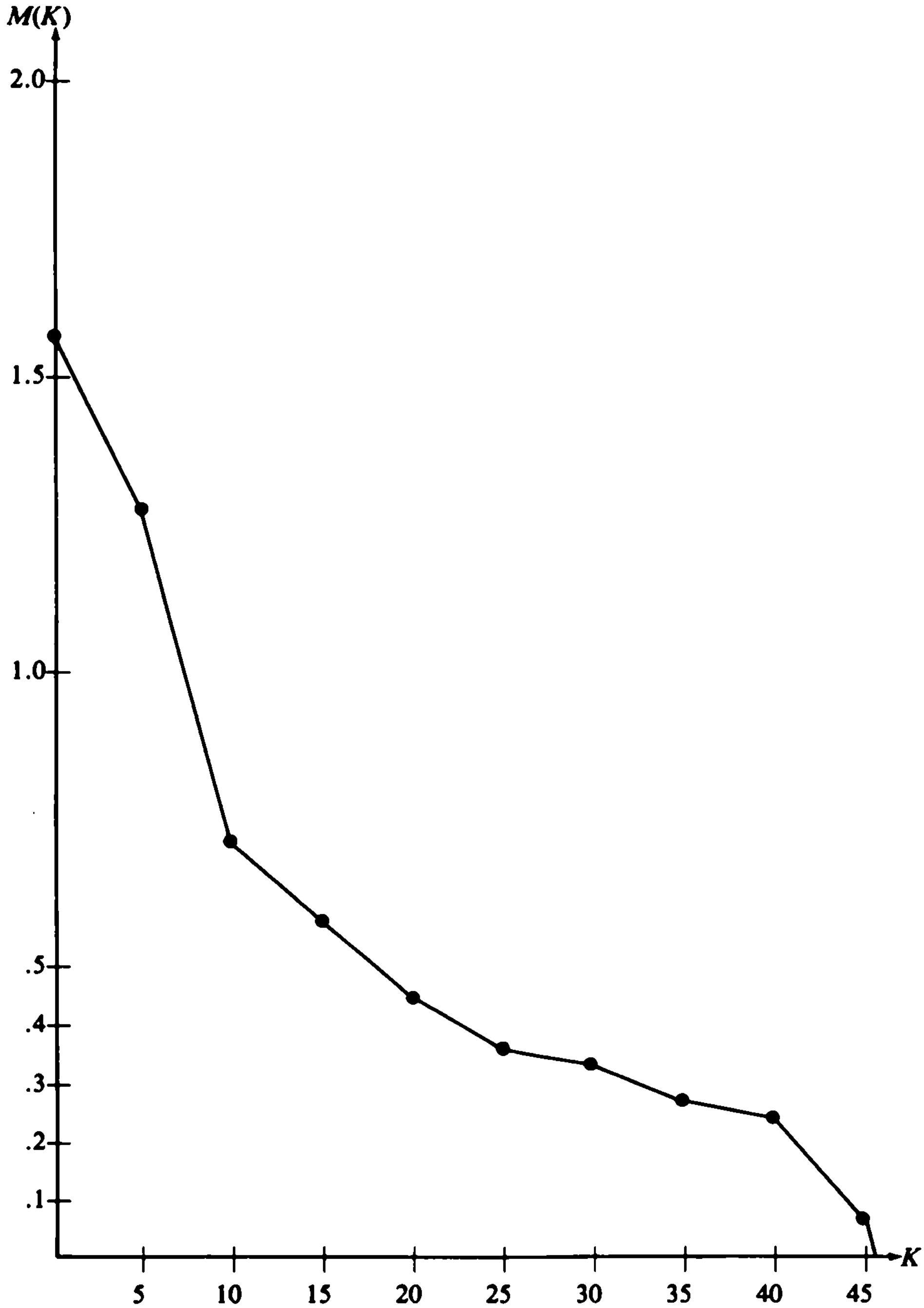
A piecewise linear curve for $M(K)$ is obtained by connecting points $(K, M(K))$ and $(K+1, M(K+1))$ for $K = 0, 1, \ldots, \bar{K}-1$ by a straight line, where $\bar{K}$ is defined to be the smallest integer such that $M(\bar{K}) \leq 0$. Although it is not true of all heuristics, in general, $M(K) \geq M(K+1)$ for all $K \geq 1$; in particular, this relationship holds for the greedy-insertion heuristic. The "least-cost solution" over all fleet sizes occurs at the largest K value for which $M(K)$ is nonnegative.

Because of management's conservatism and its desire to avoid leasing a fleet that might be too large if demand should fall, the firm sought a fleet size K^* for which $M(K^*)$ was between .2 and .3. This solution reduces the number of vehicles from what it would have been for $M(K^*)$ closer to zero. This solution is suboptimal under the given set of demand trips and also in the event that demand rises. If demand falls, however, this solution remains extremely close to optimal since some trips that would have been handled by common carrier could now be carried at a very small additional cost by the leased fleet. On the one hand, this admittedly conservative policy allows for larger deviations from optimality when demand increases, as a larger fleet size would then be optimal. On the other hand, this policy suffers only slight deviations from optimality when demand decreases, thus allowing us to generate exceptionally effective routes that included nearly all profitable trips.

The results on which the strategic fleet-size decision was made are illustrated in Figure 5. The graph in Figure 5 represents routes for a two-week period. Dealing with a two-week period was preferable to a one-week period because it is easier to work from integral numbers of trips than fractional numbers. By considering two-week periods, many trips with small average weekly demands requiring at least one trip every other week could be dealt with more easily. Based on management objectives and the shape of the curve in Figure 5, a cutoff point was agreed upon of .2 for the marginal savings rate for K vehicles. This decision suggested a leased-fleet size of 21 vehicles per week as compared to the current operating level of 30. If they leased 21 vehicles per week, the average marginal savings rate would be .58, the total savings would be $33,500 per week, and the average route time would be 114 hours (since $T = 120$ hours).

In addition to the financial savings and increased productivity information conveyed by the fleet-size determination, the firm obtained a detailed description of the routes. This information was primarily used to increase the confidence of the firm's management in the results we generated. Moreover, the information provided management with a broad view of what their operational routing configuration would look like on a week-to-week basis. In our experience, this latter phase of the study (and the close interaction it required with the firm's management personnel) was instrumental in their accepting the model's results.

FIGURE 5
Marginal Savings Rate for Leasing *K* Vehicles



GENERALIZATIONS AND ALGORITHMIC IMPROVEMENTS

As the first section indicated, we believe this class of problems is becoming increasingly important and that additional research in this area should be encouraged.

Although Algorithms 1A and 1B did not produce the best results in this case, they may prove very useful in future applications. In particular, they are based on modifications of the mathematical programming formulation given in the third section. Consequently, it may be possible to investigate the theoretical properties of the algorithms and, as a result, to improve their accuracy. By comparing their results with related relaxations to problem P, it may be possible to obtain bounds on the solution to P. Finally, our three solution procedures illustrate that different approaches to the problem may be used concurrently to overcome weaknesses that each approach may demonstrate singly. Since our mathematical program formulation cannot be solved optimally for realistic problem sizes, the ability to attack the same problem with several (complementary) heuristics is advantageous in generating near-optimal solutions.

The solutions we obtained were used in order to reach a strategic long-term decision. Since demand changes from week to week, it would be desirable to have an on-line system capable of generating a new set of routes each week. While the same algorithmic approaches would be appropriate, a variety of additional problems involving the hardware environment and data handling would have to be considered more explicitly. Moreover, the additional timing constraints described in the second section may require explicit incorporation into the model before the model can be used more generally. We were fortunate in that simple modifications allowed us to ignore the timing constraints in the original model and satisfy them after the fact. [Received: February 8, 1982. Accepted: June 15, 1982.]

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APPENDIX A
A Note on the Subtour-Breaking Constraints

This appendix demonstrates that Constraint set (5) does, in fact, prohibit the formation of illegal subtours and allows for permissible cycles.

In Figure A, two 3-node configurations with a domicile denoted by 0 are given. Constraint set (5) for vehicle k is written beside each configuration. The n_i ($i=1,2,3$) and m_j ($j=1,2,3$) come from the right-hand side of Constraint (5); they are positive numbers based on the flow variables x_{ijk} and y_{ijk}.

It should be clear that the first configuration is an illegal subtour and the second is a permissible cycle. For the first configuration, the summation of the three constraints leads to a contradiction, as desired; for the second configuration, the following f_{ijk} values are, indeed, feasible:

$$f_{01k}=0$$

$$f_{12k}=m_1$$

$$f_{23k}=m_1+m_2$$

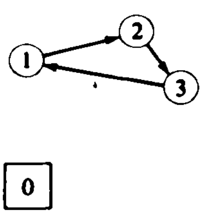
$$f_{31k}=m_1+m_2+m_3$$

$$f_{10k}=m_1+m_2+m_3.$$

In other words, Constraint set (5) behaves exactly as we require.

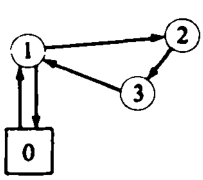
FIGURE A
Sample Configurations to Prove Efficacy of Constraint Set (5)

Configuration 1



$$\begin{aligned} f_{12k}-f_{31k}&=n_1>0\\ f_{23k}-f_{12k}&=n_2>0\\ f_{31k}-f_{23k}&=n_3>0\\ \hline 0&=n_1+n_2+n_3 \end{aligned}$$

Configuration 2



$$\begin{aligned} f_{12k}+f_{10k}-f_{01k}-f_{31k}&=m_1>0\\ f_{23k}-f_{12k}&=m_2>0\\ f_{31k}-f_{23k}&=m_3>0 \end{aligned}$$

APPENDIX B
Algorithm 1B in More Detail

This appendix treats the problem of choosing a maximum savings set of K candidate routes (no two of which cover the same node), which is a component of Algorithm 1B.

Suppose the following candidate routes emerge from the large tour displayed in Figure 3A:

1. (D1, 1, 2, D1)
2. (D1, 1, 2, 3, D1)
3. (D1, 2, 3, 4, D1)
4. (D1, 3, 4, 5, D1)
5. (D2, 4, 5, D2)
6. (D2, 4, 5, 6, D2)
7. (D2, 5, 6, 7, D2)
8. (D2, 6, 7, D2)
9. (D1, 7, 8, D1)
10. (D1, 8, 9, D1)
11. (D1, 8, 9, 1, D1)
12. (D1, 9, 1, 2, D1)

Then the set-packing problem below would result.

$$\text{Max} \sum_{j=1}^{12} q_j x_j \tag{B.1}$$

subject to

$$\begin{array}{cc} & \text{Route Number} \\ \begin{array}{c}\text{Node}\\ \text{Number}\end{array} & \begin{bmatrix} 1 & 1 & & & & & & & & & 1 & 1 \\ 1 & 1 & 1 & & & & & & & & & 1 \\ & 1 & 1 & 1 & & & & & & & & \\ & & 1 & 1 & 1 & 1 & & & & & & \\ & & & 1 & 1 & 1 & 1 & & & & & \\ & & & & & 1 & 1 & 1 & & & & \\ & & & & & & 1 & 1 & 1 & & & \\ & & & & & & & & 1 & 1 & 1 & \\ & & & & & & & & & 1 & 1 & 1 \end{bmatrix} \end{array} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{12} \end{bmatrix} \le \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$\sum_{j=1}^{12} x_j \le K \qquad x_j = 0 \text{ or } 1 \text{ for all } j.$$

Generalizing the problem of choosing the maximum savings set of routes may be formulated as a mathematical program of the form: maximize qx subject to $Fx \le \mathbf{1}$, $\mathbf{1}x \le K$ and x binary where F is a (0,1) matrix, and $\mathbf{1}$ is a vector of ones. The columns of F correspond to routes and the rows to nodes in the TSP tour (see [3]). The jth component of $q(q_j)$ is the savings associated with the jth candidate route.

We explain the set-packing approach in more detail by once again referring to Figure 3. In Figure 3 a traveling salesman tour over all nodes (excluding domicile nodes) is found. From this large tour, candiate routes are generated consisting of a sequence of nodes contiguous in the large traveling salesman tour linked at both ends to the most convenient (nearest) domicile. For a route to qualify as a candidate route, its duration must not exceed T. Three candidate routes are displayed in Figure 3B. In this appendix, we list a total of 12 candidate routes from which the corresponding mathematical program is formulated in Equation (B.1). If we let $K=3$, then we see that $x_3=x_7=x_{11}=1$ is a feasible solution with total savings $q_3+q_7+q_{11}$. The optimal solution to this problem selects the set of K or fewer routes that results in the greatest overall savings.

Since each candidate route contains consecutive nodes from the TSP tour, the matrix F has the "consecutive-ones" property with "wrap around." Matrices with the consecutive-ones property are totally unimodular [7]. This particular mathematical program, which is referred to as a set packing problem with special structure, can be solved even more efficiently, however, by fully exploiting the consecutive-ones property [13]. The resultant problem is a constrained maximum-weight path problem on an acyclic network.

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